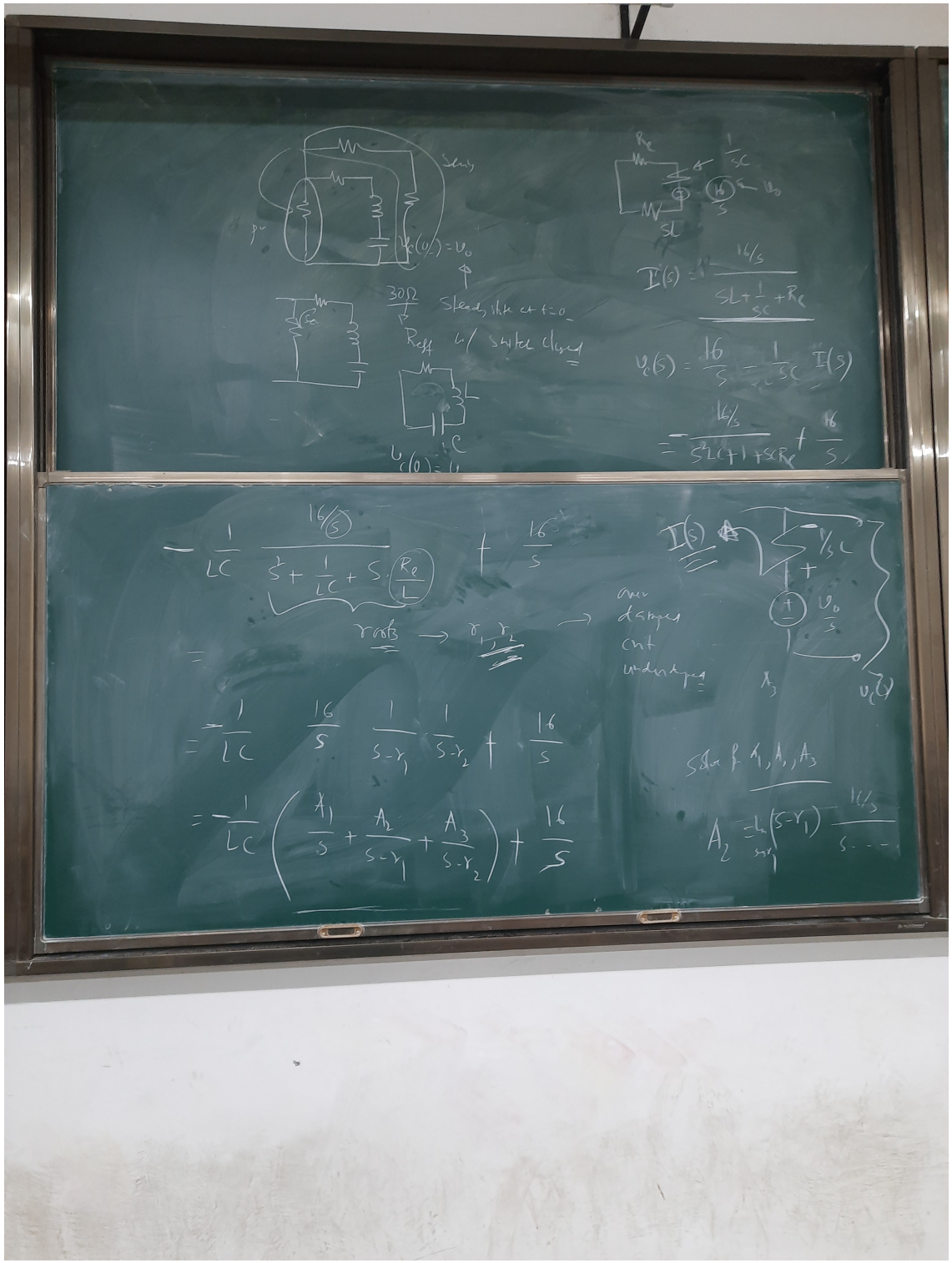
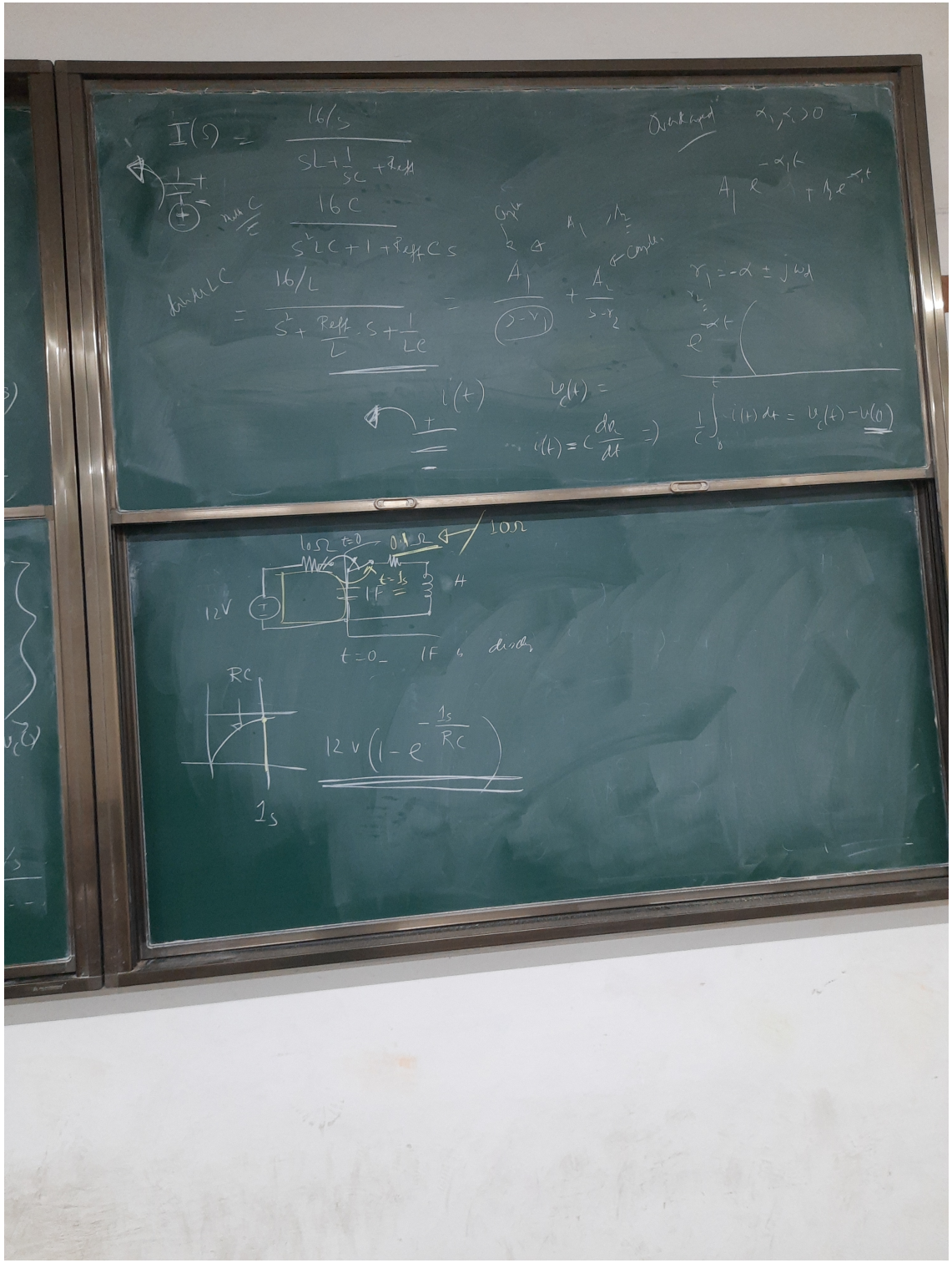

30Ω
Steady state at t=0⁻
R_{eff} w/ switch closed
$I(s) = \frac{16/s}{sL + \frac{1}{sC} + R_0}$
$V_c(s) = \frac{16}{s} - \frac{1}{sC} I(s)$
$= \frac{16/s}{s^2LC + 1 + sCR_0} + \frac{16}{s}$
$= -\frac{1}{LC} \frac{16/s}{s^2 + \frac{1}{LC} + s\left(\frac{R_0}{L}\right)} + \frac{16}{s}$
roots → r_1, r_2
over damped
crit
underdamped
$= -\frac{1}{LC} \left(\frac{A_1}{s} + \frac{A_2}{s - r_1} + \frac{A_3}{s - r_2} \right) + \frac{16}{s}$
Solve for A_1, A_2, A_3
$A_2 = \lim_{s \to r_1} (s - r_1) \frac{16/s}{s(s - r_1)(s - r_2)}$


I(s) =
16/s
sL + 1/sC + Reff
16C
s²LC + 1 + ReffCs
16/L
s² + Reff/L · s + 1/LC
Overdamped
α₁, α₂ > 0
A₁ e^(−α₁t) + A₂ e^(−α₂t)
A₁/(s−r₁) + A₂/(s−r₂)
r₁ = −α ± jωd
e^(−αt)
i(t)
v_c(t) =
i(t) = C dv/dt
1/C ∫₀ i(t) dt = v_c(t) − v(0)
10Ω t=0
0.1Ω
10Ω
12V
t=1s
1F
H
t=0₋ 1F is discharged
RC
1s
12 V (1 − e^(−1s/RC))